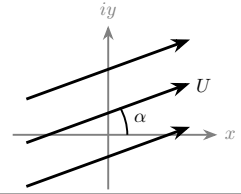
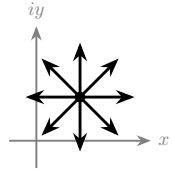
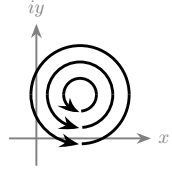
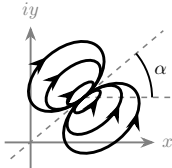
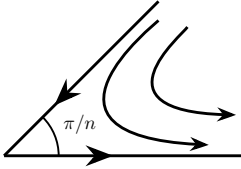
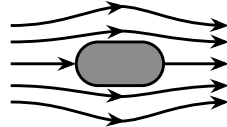
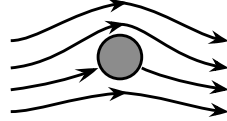


A Brief Handbook of Elementary Ideal Flows

Elementary flow	Complex potential, $F = \phi + i\psi$	Potential, $\phi(x, y)$ or $\phi(r, \theta)$	Streamfunction, $\psi(x, y)$ or $\psi(r, \theta)$	Sketch of streamlines
① Uniform flow at angle α to the horizontal U is real but A is complex	Az with $A = Ue^{-i\alpha}$ U is real but A is complex	$U(x \cos \alpha + y \sin \alpha)$ or $Ur \cos(\theta - \alpha)$	$U(y \cos \alpha - x \sin \alpha)$ or $Ur \sin(\theta - \alpha)$	
② Source ($Q > 0$; for sink replace Q by $-Q$) located at $z_0 = x_0 + iy_0$	$\frac{Q}{2\pi} \log(z - z_0)$	$\frac{Q}{2\pi} \ln \sqrt{(x - x_0)^2 + (y - y_0)^2}$ or, for $z_0 = 0$, $\frac{Q}{2\pi} \ln r$	$\frac{Q}{2\pi} \tan^{-1} \left(\frac{y - y_0}{x - x_0} \right)$ or, for $z_0 = 0$, $\frac{Q}{2\pi} \theta$	
③ Vortex (counterclockwise for $\Gamma > 0$) located at $z_0 = x_0 + iy_0$	$\frac{-i\Gamma}{2\pi} \log(z - z_0)$	$\frac{\Gamma}{2\pi} \tan^{-1} \left(\frac{y - y_0}{x - x_0} \right)$ or, for $z_0 = 0$, $\frac{\Gamma}{2\pi} \theta$	$-\frac{\Gamma}{2\pi} \ln \sqrt{(x - x_0)^2 + (y - y_0)^2}$ or, for $z_0 = 0$, $-\frac{\Gamma}{2\pi} \ln r$	
④ Doublet located at $z_0 = x_0 + iy_0$ and rotated by angle α	$\frac{me^{i\alpha}}{z - z_0}$ m is real	$\frac{m \cos \left(\tan^{-1} \left(\frac{y - y_0}{x - x_0} \right) - \alpha \right)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}$ or, for $z_0 = 0$, $\frac{m \cos(\theta - \alpha)}{r}$	$-\frac{m \sin \left(\tan^{-1} \left(\frac{y - y_0}{x - x_0} \right) - \alpha \right)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}$ or, for $z_0 = 0$, $-\frac{m \sin(\theta - \alpha)}{r}$	
⑤ Flow in a corner of angle π/n ($n > 1/2$) $1 < n < 2$: flows past wedges of angle $2\pi \frac{n-1}{n}$ $n = 2$: stagnation-point flow onto a horizontal plate	Az^n	[inconvenient to use Cartesian coords] or $Ur^n \cos(n\theta)$	[inconvenient to use Cartesian coords] or $Ur^n \sin(n\theta)$	
⑥ Flow past a Rankine body (a is a real number/horizontal shift) ① $\{\alpha = 0\}$ + ② $\{z_0 = -a\}$ - ③ $\{z_0 = a\}$	$Uz + \frac{Q}{2\pi} \log \left(\frac{z + a}{z - a} \right)$	$Ux + \frac{Q}{2\pi} \ln \sqrt{\frac{(x + a)^2 + y^2}{(x - a)^2 + y^2}}$ [inconvenient to use polar coords]	$Uy + \frac{Q}{2\pi} \left[\tan^{-1} \left(\frac{y}{x + a} \right) - \tan^{-1} \left(\frac{y}{x - a} \right) \right]$ [inconvenient to use polar coords]	
⑦ Flow past a cylinder with circulation ① $\{\alpha = 0\}$ + ④ $\{\alpha = 0, m = UR^2, z_0 = 0\}$ - ③ $\{z_0 = 0\}$	$U \left(z + \frac{R^2}{z} \right) + \frac{i\Gamma}{2\pi} \log z$	$U \left(1 + \frac{R^2}{x^2 + y^2} \right) x - \frac{\Gamma}{2\pi} \tan^{-1} \left(\frac{y}{x} \right)$ $U \left(1 + \frac{R^2}{r^2} \right) r \cos \theta - \frac{\Gamma}{2\pi} \theta$	$U \left(1 - \frac{R^2}{x^2 + y^2} \right) y + \frac{\Gamma}{2\pi} \ln \sqrt{x^2 + y^2}$ $U \left(1 - \frac{R^2}{r^2} \right) r \sin \theta + \frac{\Gamma}{2\pi} \ln r$	

Inspired by Tables 12.1 and 12.2 of Granger's *Fluid Mechanics* (Dover Publications, 2012). Flow ① – ⑤ are the “building blocks” and the rest are superposition examples.